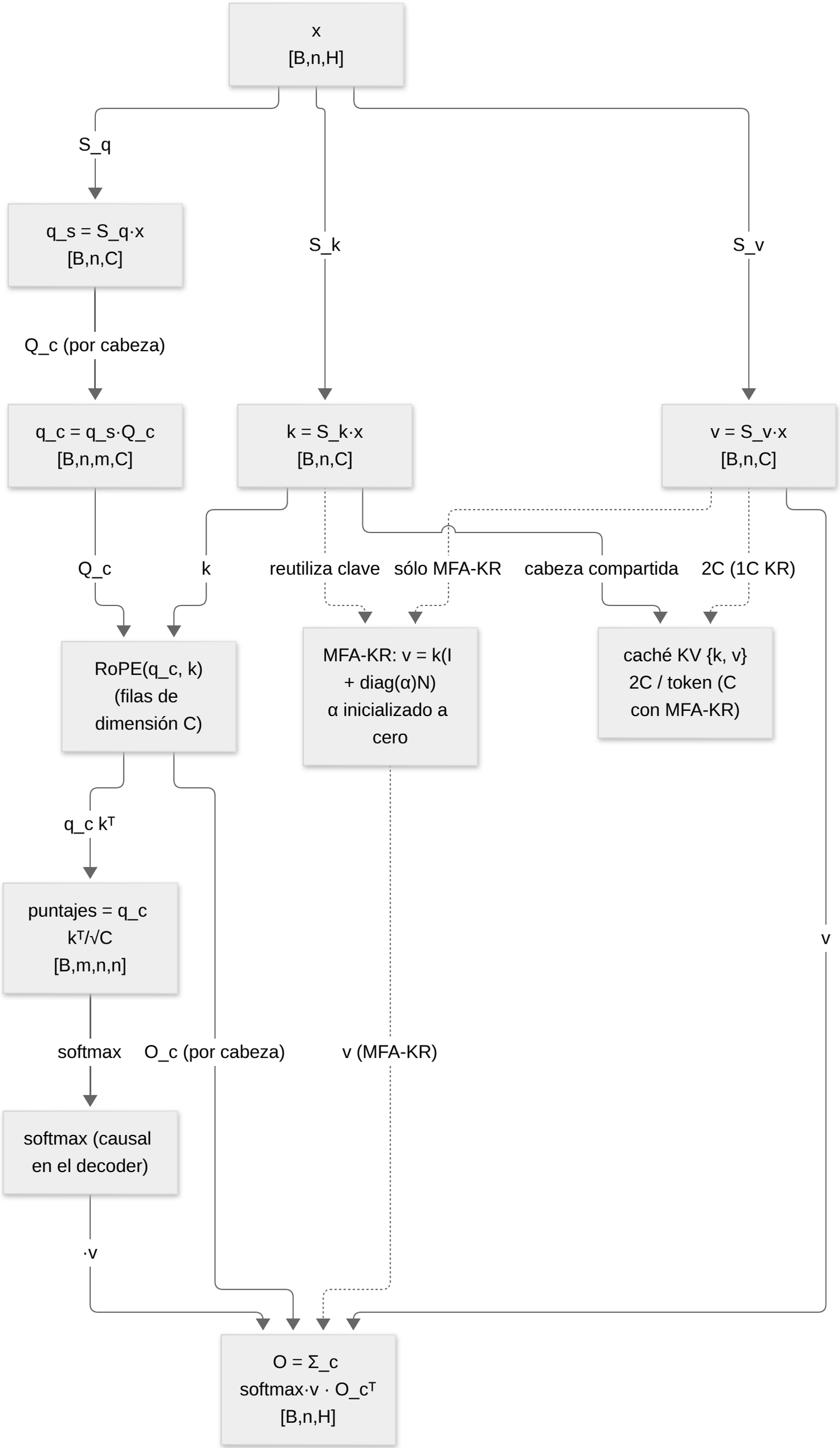

x
[B,n,H]
S_q
S_k
S_v
q_s = S_q·x
[B,n,C]
Q_c (por cabeza)
q_c = q_s·Q_c
[B,n,m,C]
k = S_k·x
[B,n,C]
v = S_v·x
[B,n,C]
Q_c
k
reutiliza clave
sólo MFA-KR
cabeza compartida
2C (1C KR)
RoPE(q_c, k)
(filas de
dimensión C)
MFA-KR: v = k(I
+ diag(α)N)
α inicializado a
cero
caché KV {k, v}
2C / token (C
con MFA-KR)
q_c k^T
puntajes = q_c
k^T/√C
[B,m,n,n]
softmax
O_c (por cabeza)
v (MFA-KR)
v
softmax (causal
en el decoder)
·v
O = Σ_c
softmax·v · O_c^T
[B,n,H]